



Document Management System



Sultanate of Oman

Ministry of Health

The Royal Hospital

Department of Child Health

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Title: Conscious Sedation for Paediatric

1. Purpose

The aim of the policy is to assure a consistent process that promotes safe and effective sedation for children undergoing procedures.

Conscious sedation is the process of giving a sedative, hypnotic, or analgesic drug by any route, which may result in partial loss of consciousness and/or protective reflexes of their airways, in order to facilitate a diagnostic or therapeutic procedure. As a result these patients require close monitoring of their cardiopulmonary status.

2. Goal

- 2.1. Guard the patient's safety and welfare
- 2.2. Minimize physical discomfort and pain
- 2.3. Control anxiety, minimize psychological trauma, and maximize the potential for amnesia
- 2.4. Control behavior and/or movement to allow the safe completion of the procedure
- 2.5. Return the patient to a state in which he/she can be safely discharged

3. Indications

- 3.1. Painless procedures
 - 3.1.1. Transthoracic ECHO
 - 3.1.2. Diagnostic Imaging (e.g CT, MRI)
 - 3.1.3. Electrophysiology studies (e.g EEG, ECG, brainstem evoked responses ...etc.)
 - 3.1.4. Radiotherapy
- 3.2. Painful procedures

3.2.2. ICDInsertion

3.2.3. Centralvenous catheterization and peripheral inserted central line (PICC)

3.2.4. Lumbarpuncture and intrathecal injections

3.3. Anyprocedure in which the use of sedation/analgesia in a non-intubated patient isperformed

4. Rolesand Responsibilities

4.1. Patient record must be available at the time ofsedation

4.2. The physician performing procedure prior to theprocedure must obtain a written consent

4.3. Sedation to be administered by a senior nursecertified in IV with the presence of a doctor, expert in airway management andPALS certified

4.4. The sedation must be prescribed in Al-Shifa, checkedwith the doctor for the dose, preparation and dilution

4.5. A registered nurse should be responsible forassessment and monitoring of thepatient pre, during and post procedure

4.6. Conscious sedation form should be used todocument the details of the patient throughout the process

5. Pre-sedation Assessment

A pre-sedation medical assessment of suitability for sedation should be

performedand documented. This includes a historyand examination as well as looking at contraindications.

5.1. PhysiciansResponsibility

5.2.1. History:

- a. Concurrent medicalillnesses – respiratory illness, volume depletion, fever
- b. Medications
- c. Allergies orreaction prior to previous exposure to sedation
- d. Prior adversereactions to anesthetic/ sedative agents (e.g. paradoxical reactions, nausea,vomiting, airway difficulties)
- e. History of sleepdisordered breathing or snoring
- f. Major medicalillnesses, physical abnormalities, and neurologic problems
- g. Last solid/ liquidintake

5.2.1. PhysicalExamination:

- a. Airway examination
- b. Cardiorespiratoryexamination

5.1.3. Laboratory : shouldcheck relevant investigation results

5.1.4. Physicians should becomepetent in:

- a. Pediatric airwaymanagement and resuscitation
- b. Patient selection,contraindications & preparation

- d. Recognition and treatment of the complications associated with the medications to be used

5.2. Nursing Responsibility

5.2.1. History:

- a. Should ask about allergies

5.2.2. Physical examination

- a. Weight should be measured and documented for drug dosing
- b. Should check and document baseline arousal score, respiratory rate, heart rate, blood pressure and oxygen saturation prior to the procedure.

5.2.3. Staff nurses should be:

- a. Aware of the basic airway management and resuscitation
- b. Knowledgeable of patient preparation and monitoring procedures
- c. Familiar with proper documentation of the conscious sedation monitoring form

6. Risk Assessment

The following categories are considered high risk:

- 6.1. Vomiting, bowel obstruction, GE reflux
- 6.2. Asthma exacerbation, pneumonia
- 6.3. Cardiac disease, Hypovolemia, Sepsis
- 6.4. Altered mental status, neurologic/neuromuscular disorder
- 6.5. History of sedation failure
- 6.6. Age <1 year
- 6.7. Severe obesity
- 6.8. Moderate or severe systemic disease which limits activity

7. Exclusion criteria

If the following are present, anesthesia/intensive care should be consulted for the conscious sedation

- 7.1. Difficult airway syndromes – abnormal face, mouth, big tongue, dentition or neck
- 7.2. Sleep apnea, stridor, airway obstruction, severe asthma
- 7.3. Tracheal abnormalities
- 7.4. Severe cardiorespiratory disease
- 7.5. Head injury and altered consciousness
- 7.6. Raised intracranial pressure
- 7.7. Severe neurological impairment and/or bulbar dysfunction

7.9. Malignant hyperthermia

7.10. Known allergy or adverse reaction to sedative

8. Fasting guideline

Keeping the child fasting according to the protocol prior to conscious sedation is recommended.

9. The aims of fasting:

9.1. To minimize the risk of aspiration of gastrointestinal contents.

9.2. To maximize patient comfort in the sedation phase.

9.3. To minimize any potential significant physiological changes that may occur from prolonged restriction of oral intake.

Ingested material	Minimum fasting period (hours)
Clear liquids: water, fruit juices without pulp, carbonated beverages, clear tea, black coffee	2
Breast milk	4
Infant formula, nonhuman milk	6
Light meal: a light meal typically consists of toast and clear liquid.	6
Full meals: Meals that include fried or fatty food such as meat may prolong gastric emptying time.	8

10. Equipment

It is helpful to use an acronym that allows the same setup and checklist for every procedure. A commonly used acronym useful in planning and preparation for a procedure is **SOAPME**:

10.1. **S** = Size-appropriate **Suction** catheters and a functioning **suction** apparatus (eg, Yankauer-type suction)

10.2. **O** = An adequate **Oxygen** supply and functioning flow meters/other devices to allow its delivery Oxygen flow meter, oxygen extension tubing and age appropriate masks.

10.3. **A** = **Airway**: Be prepared for emergency intubation if required. Keep the following ready:

10.3.1. Oral and Nasal Airways

10.3.2. Laryngoscope handles (tested) with different size blades (bulbs tested)

10.3.3. Appropriate size Endotracheal Tubes,

10.3.4. Stylets, appropriately sized for endotracheal tubes

10.3.5. Nasogastric tube

10.6. **M = Monitors:** functioning pulse oximeter with size-appropriate oximeter probes and other monitors as appropriate for the procedure (eg, noninvasive blood pressure, end-tidal carbon dioxide, ECG, stethoscope)

10.7. **E = Special equipment or drugs** for a particular case (e.g. defibrillator, Propranolol for cyanotic spell in TOF... etc.)

10.8. Cardio respiratory monitor with age appropriate cable and electrodes with appropriate size blood pressure cuffs and pulse oximeter.

10.9. Well-equipped emergency trolley with defibrillator, appropriate bag mask valve BVM device with appropriate size mask.

11. Continuum of depth of sedation

	Minimal Sedation (Anxiolysis)	Moderate sedation (Conscious sedation)	Deep Sedation/ Analgesia	General Anaesthesia
Responsiveness	Normal response to verbal stimulation	Purposeful response to verbal or tactile stimulation	Purposeful response following repeated or painful stimulation	Un-arousable even with painful stimuli
Airway	Unaffected	No intervention required	Intervention may be required	Intervention often required
Spontaneous ventilation	Unaffected	Adequate	May be inadequate	Frequently inadequate
Cardiovascular function	Unaffected	Usually maintained	Usually maintained	May be impaired

Note: "if the level of sedation is more than moderate, intubation may be needed so alert PICU".

12. Procedure

12.1. Confirm patient ID and review risks and benefits of the procedure.

12.2. Take history and perform physical examination aiming at determining baseline status of the patient and identifying factors that may increase the patient's risk during the period of sedation.

12.3. Review NPO time, risk assessment and exclusion criteria as outlined in the Procedural Sedation Template. **If any is identified then anesthesia or intensive care consultant is to be consulted.**

12.4. Explain to the parents if appropriate, prior to administration of sedative medication regarding the risks and potential adverse effects of sedation, anticipated sedative effects, reason for sedation and potential options other than sedation.

12.5. Take informed consent and document in the patient record.

- 12.7. Assemble and check the function of all equipment and supplies.
- 12.8. Check for the availability of appropriate BMV masks sizes.
- 12.9. Ensure that all alarms are set appropriately per patient's age and history.
- 12.10. Monitor the patient's airway and vital signs and pain management throughout the procedure. Nursing performs this function in conjunction with the physician
- 12.11. A physician certified for the management of airway must be immediately available at all time.
- 12.12. Apply supplemental oxygen when indicated.
- 12.13. Document pre-procedure vital signs as per the recording form attached.
- 12.14. Assign a staff nurse, who will be present from the time of administration of sedative drugs until the child returns to baseline status. She/he shall continuously assess the patient and document assessments and findings as outlined on the procedural sedation template.

13. The possible side effects of procedural sedation

- 13.1. **O2 desaturation:** Stimulate patient, instruct to take deep breaths. Give O2
- 13.2. **Respiratory depression/apnea:** Breathe frequently, give 100% O2, REVERSAL AGENTS
- 13.3. **Symptomatic Bradycardia:** Give atropine (especially if fentanyl or morphine used)
- 13.4. **Hypotension :** Fluid bolus
- 13.5. **Hemodynamic instability -> CALL A CODE!!**

14. Post procedure

- 14.1. Monitoring should be continued at least until patient is back to his pre sedation status.
- 14.2. No further sedation will be given
- 14.3. Oxygen saturation is greater than 95% in room air
- 14.4. Dispose and clean equipment according to policies and procedures.
- 14.5. Document the initiation of or changes of patient's condition during procedure in the sedation checklist.
- 14.6. Documentation should include a full assessment of the patient during and after procedure as per the hospital policy

15. Discharge criteria should include

- 15.1. Back to the pre-procedure mental status and aroused easily
- 15.2. Ability to ambulate (if appropriate) or **able to sit without support**
- 15.3. Ability to verbalize appropriate to age. Oriented to caregiver.
- 15.4. Final set of vital signs are within normal limits for the child's age
- 15.5. 30 minutes from administration of IV medication or two hours after antidote (if given)

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
Clinical Practice Guideline for Oral Conscious Sedation of Children Being Performed By Non Anaesthetic Personnel	Princess Margaret hospital for children	September 2011	1-6
Guideline for Pediatric Procedural Sedation and Analgesia in the Emergency department/ Annals of Emergency Medicine	Scott D.et all	2005	177-196
Pediatric Sedation Outside of the Operating Room: A Multispecialty International Collaboration/AAP	Joseph P. Cravero	2015	17-31
Administration of Medications for Sedation/Analgesia	The Massachusetts Board of Registration in Nursing	9/11/13	-

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